HEXAGON

Brochure

MSC solutions for aerospace suppliers

Delivering certainty in a changing business landscape

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Opportunities and challenges

Increasing globalization of the aerospace industry is presenting the supplier community with new opportunities and challenges as the emerging economies are changing the business landscape.

- To reduce costs and improve efficiency, OEMs (Original Equipment Manufacturers) are turning to a smaller group of trusted worldwide suppliers
- To maintain competitiveness, suppliers are seeing a strong need to adopt design and analysis solutions used by OEMs
- With increased product development responsibilities, suppliers are seeing the need for more efficient communication channels and streamlined data sharing

MSC Software, the original developer of the highly trusted simulation solutions used by aerospace industry, is ideally positioned to partner with supply chain in the journey to improve product efficiency and reduce costs.

In addition to providing suppliers with industry standard and proven simulation solutions across the broad spectrum of Computer Aided Engineering (CAE), MSC enables customers to maintain uniformity and reliable information sharing across departments and the supply chain.

Trusted solution

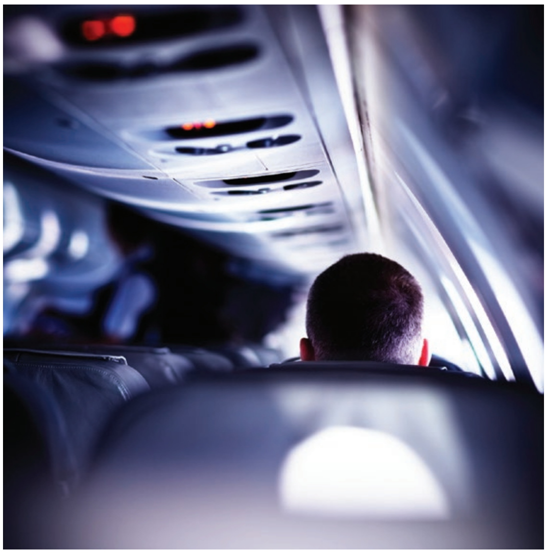
MSC Software's solutions are the most trusted by the aerospace OEMs, especially since MSC Nastran was first developed to help address the design challenges faced by NASA and the aerospace industry in the 1960's.

MSC's products are used by all the major OEMs to understand current and new designs, study failures and improve their products.

MSC partners and collaborates with key players of the industry to ensure timely and effective solutions are delivered.

Specific technologies are developed for the aerospace industry by people with extensive experience in aerospace engineering. You can rely on products of MSC Software to address your specific needs in:

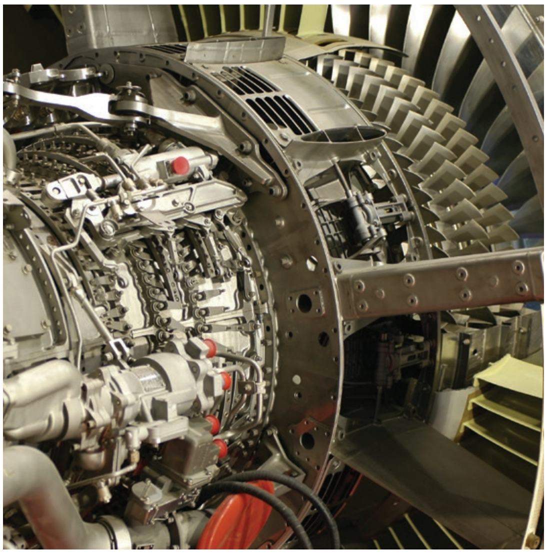
- Loads
- Dynamics
- Noise and vibration
- Durability
- Damage and failure
- Structural performance
- Thermal analysis
- Flight control systems
- Mechanical systems
- Stress analysis



Higher confidence with simulation

Use MSC's proven simulation solutions to complement and minimize physical testing and gain early insight into your product performance. We help you:

- Accurately determine loads from static and dynamic aeroelastic simulations
- Compute structural static, dynamic and thermal response of the components and assemblies
- Analyze interaction with neighboring components of the assembly
- Obtain meaningful sizing and margin calculations to prevent unintentional failures
- Benefit from uninterrupted path from external loads calculations to internal loads and downstream sizing and margin calculations
- Map temperature, load and deformation between simulation models and disciplines for improved accuracy



So, whether your organization has a single or multi-disciplinary focus, MSC's product suite enables simulation based design and validation decisions along the entire loads to certification process.

Robust and efficient aeroelastic and internal loads analysis

Knowledge of static and dynamic flight and ground loads, as well as flutter characteristics across the flight envelope is critical to aircraft structural design. An accurate aeroelastic loads process is also a necessity for efficient margins and downstream certification analyses.
With MSC's solution, you can:

- Sample the flight envelope at hundreds of thousands of points in the sky and on the ground, giving you a competitive advantage
- Gain improved efficiency in aircraft development with efficient loads computation as downstream sizing and margin simulations require the loads as inputs
- Benefit from MSC's unique and unmatched industrial experience in bringing vehicle level loads and dynamics technologies to market, that can also be tailored to leverage and integrate with your existing processes, increasing your productivity

Powerful stress analysis solution for virtual product development

Accurate prediction of force, stress and strain margins of all structural components is crucial in order to achieve the goal of weight minimization and material savings. MSC provides a single, integrated platform to address the multidisciplinary loading scenarios that need to be analyzed, which include:

- Linear stress analysis with contact
- Linear and nonlinear dynamic analysis
- Nonlinear stress analysis to include all nonlinearities, including materials, and large deformations and strains

With comprehensive stress analysis capabilities, you can:

- Easily model and simulate interaction between multiple components of an assembly through contact body modeling
- Accurately simulate multitude of nonlinear materials, including metals, elastomers, plastics, shape memory alloys, composites and other customized materials

Understand interaction between multiple physical phenomena through coupled analyses like fluid-structure interaction, thermomechanical anlaysis and structural acoustic studies.

By using MSC's reliable simulations to study performance of your products in harsh environmental conditions, you can extract all the information you need to make design decisions with more certainty.

Failure prediction with more confidence

Failure of aerospace structural components can lead to catastrophic outcomes, and so prediction plays a critical role in ensuring prevention. With MSC's solutions, you gain significant advantage in product development by your improved knowledge of failures.

- Study complex structural failures at a lower cost and in a safer environment for your engineers
- Gain access to robust, comprehensive failure technologies in fracture mechanics, progressive failure, damage mechanics and life-to-failure solutions
- Use the industry leading fatigue solution, perform durability analysis to address product performance and failure issues early in the product development cycle
- Gain better understanding of crack propagation path and rate, helping you avoid failures in service

Rotor dynamic stability for improved engine performance

Rotating machinery poses complex and unique design challenges. Stability and vibration control/minimization of aircraft rotor systems (propellers, jet engines etc.) is critical for performance, noise and vibration and safety of aircraft systems. MSC's rotor dynamic analysis methods, developed in collaboration with NASA, leading OEMs and a worldwide rotordynamics consortium, provide robust procedures to:

- Analyze and optimize rotating systems
- Detect product design flaws
- Determine system instabilities
- Design bearings to eliminate noise and vibration as well as calculate safe operating ranges and conditions before the products go into service

Lighter and better composites

The aerospace industry is a pioneer in the use of modern composite materials and the use of composites in aircrafts has been steadily growing over the past few decades. MSC provides a comprehensive suite of solutions to address engineering, manufacturing and structural analysis requirements, which include:

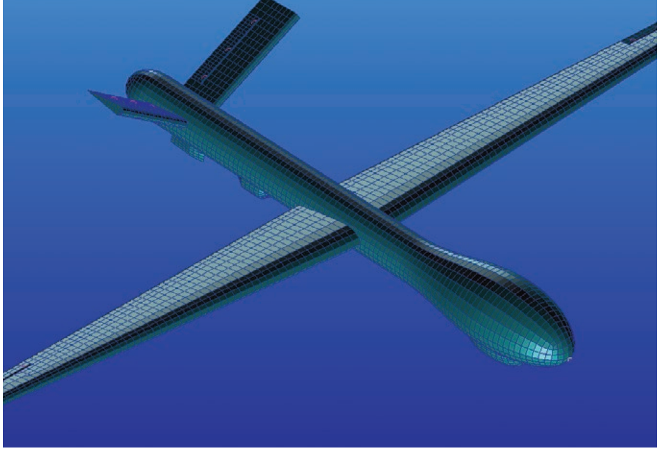
- Stress and strain response to mechanical and thermal loads
- Damage build up and failure
- Manufacturing methods and curing process parameters
- Variability of the manufactured materials and components
- Optimal layup and sizing for structural efficiency and design robustness

Using MSC's solutions, you can more confidently keep up with changing OEM requirements and schedules while reducing the costs of physical testing as simulation methodologies are validated and deployed.

Quieter aircraft forpassenger comfort and eco-friendliness

Passenger comfort is a competitive advantage and a critical criterion of aircraft design, and noise and vibration control is a major contributor to the comfort. Engine vibration, which is a dominant noise source, needs to be addressed early in the design process. At the same time, ambient aero-acoustic noise levels from the engines and flight structures need to addressed to meet requirements of various national and local regulatory agencies.

MSC's solutions are consistently proven to be superior and are routinely used for noise and vibration simulations in both aircraft and automobiles. You can take advantage of this widely used technology as you work with new designs and materials to improve the value you provide to the OEMs.



Improved safety

Safety is of utmost concern in the design of any aircraft. It is not just the performance of each component that matters, but the efficacy and robustness of a complete aircraft under adverse conditions that demands further study.

MSC has been providing technologies, such as the following, to OEMs to address various safety concerns. this widely used technology as you work with new designs and materials to improve the value you provide to the OEMs.

- Crash landing/drop test
- Water landing/ditching
- Hydroplaning and skidding on runway
- Bird strike
- Store ejection
- Dummy analysis

So, whether you manufacture seats, landing gears, engines, interiors or airframe components and assemblies, MSC offers you superior solutions to address any safety concerns. Since these tests are some of the most expensive ones, you gain significant returns on your testing and engineering investments by leveraging MSC's products in the design and analysis process.

Flight and system controls

Flight control systems include hydraulics, pneumatics, electrical systems and other electro-mechanical systems that enable pilots to control the various operating mechanisms. With MSC's solutions you can:

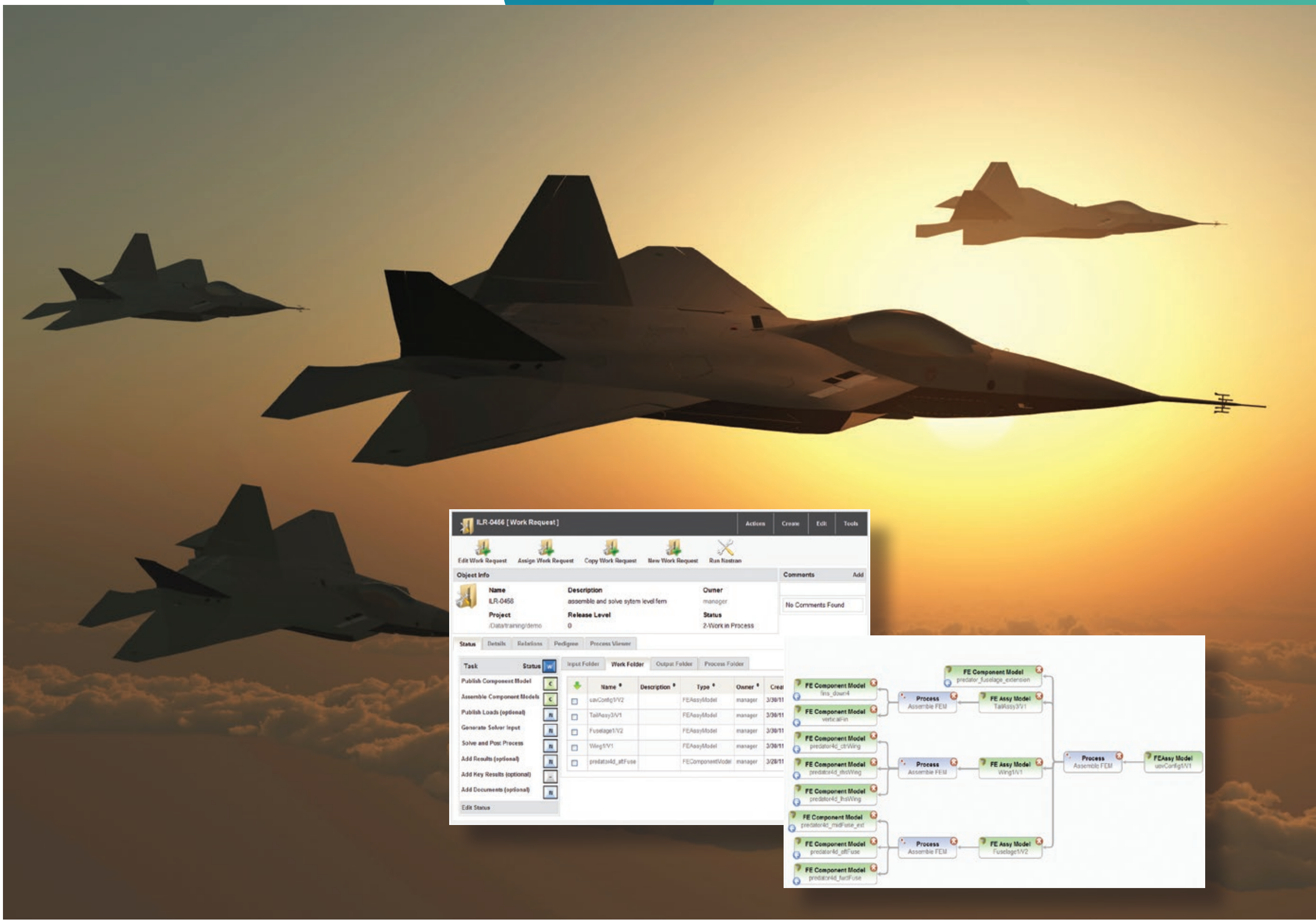
- Analyze the aircraft systems and controls software along with multibody dynamic and FEA technologies
- Gain insight into loads and dynamic system events with simulations integrated with open or closed loop controls
- Improve efficiency, safety and ride comfort through robust and well-tuned control systems
- Make design decisions confidently with integrated modeling of control systems into structural and dynamic and environmental system models

Innovation through design optimization

Material savings, weight reduction, performance improvements and manufacturability all directly affect the economic viability and competitiveness of an aircraft. You can foster innovation and develop robust designs by making effective use of MSC's optimization solutions such as:

- Design sensitivity analysis
- Automatic optimization methods to achieve weight and performance objectives while satisfying certification stress/fatigue/performance criteria
- Integration of proprietary design criteria, margins, manufacturing and performance constraints
- Multi-disciplinary optimization simulations
- Multi-model optimization (MMO) to integrate models from different disciplines, and simultaneously optimize the complete design for strength, dynamics, flight loads performance and flutter

By making effective use of optimization solutions, you can enhance your competitive advantages through innovation while keeping the development costs down.



Flexible solution that adapts to your needs

With MSC's flexible, token based licensing system, you gain access to the breadth and depth of MSC's world class simulation solution portfolio.

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Benefits:

- Maximize your productivity and stretch your engineering budget by allowing you to use whatever simulation tools you want, whenever you need them
- Reduce the need to purchase products for specific tasks and gain flexibility opening up new simulation possibilities

Simulation process management for improved productivity

You have a simulation process unique to your organization and valuable data accumulated during design and product development phases. If these massive amounts and various types of simulation data are not managed effectively, it could result in poor re-use of knowledge as well as communication and process inefficiencies within the company.

Simulation process and data management solutions from MSC offer you several competitive advantages with which you can:

- Ensure that you retain your intellectual property in an efficient manner
- Improve credibility of simulations by having complete audit trail
- Improve sharing and collaboration across teams
- Securely store and protect your simulation data
- Enhance communication with OEMs and other partners as you work through your design iterations and virtual testing

Engineering services expertise you can trust

In addition to offering multidiscipline analysis technologies to meet your challenging simulation needs, MSC also has a global engineering services team to help you get the most out of your investments. Through services like training and mentoring, helping with your work overflow, and simulation process capture and automation, our experts can help you realize the benefits of Virtual Product Development (VPD) rapidly and effectively.

HEXAGON

Hexagon is a global leader in digital reality solutions, combining sensor, software and autonomous technologies. We are putting data to work to boost efficiency, productivity, quality and safety across industrial, manufacturing, infrastructure, public sector, and mobility applications.

Our technologies are shaping production and people-related ecosystems to become increasingly connected and autonomous – ensuring a scalable, sustainable future.

Hexagon's Manufacturing Intelligence division provides solutions that use data from design and engineering, production and metrology to make manufacturing smarter. For more information, visit hexagonmi.com.

Learn more about Hexagon (Nasdaq Stockholm: HEXA B) at hexagon.com and follow us @HexagonAB.